

Mason Bott

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Education

University of Colorado Boulder

Boulder, CO

Master of Science in Computer Science, Artificial Intelligence Concentration

Expected May 2027

Bachelor of Arts in Computer Science, Minor in Mathematics (GPA: 3.96)

Dec 2025

- **Activities:** Competitive Programming & Cybersecurity Club, HackCU Hackathons
- **Relevant Coursework:** Algorithms, Data Structures, Database Systems, Data Science, Operating Systems, Machine Learning, Deep Learning, Reinforcement Learning

Experience

Autonomous Robotics & Perception Group, CU Boulder

Boulder, CO

Software Engineering & AI Researcher

June 2025 – August 2025

- Collaborated with graduate researchers and a peer student to engineer an AI reasoning agent using LangGraph, LLMs, and RAG that translates semantic, spatial, and temporal queries into actionable searches over a database.
- Designed a multi-vector spatial memory pipeline with hybrid routing, allowing proximity retrieval to bypass hierarchical graph bottlenecks for granular object searches accelerating retrieval speeds by **3x**.
- Architected a data ingestion pipeline utilizing Vision-Language Models (VLMs) and custom filtering algorithms, reducing stored video data by **95%** while preserving critical context.
- Built a Kotlin Android app with Google ARCore for real-time pose estimation, reliably streaming live tracking telemetry to a Boston Dynamics Spot robot via ROS.

Projects

Vantablock | *Rust, WGPU, WGSL, WebGPU, Nix*

- Engineered a high-performance 3D voxel game engine in Rust with WGPU, architecting a modular multi-pass graphics pipeline supporting opaque, transparent, skybox, and shadow passes.
- Optimized GPU memory bandwidth by designing custom WGSL shaders utilizing programmable vertex pulling and bit-packed face attributes, unpacking geometry on-chip to minimize VRAM transfer overhead by **86%**.
- Implemented directional sun shadow mapping with depth projection sampling and texture atlas rendering to enhance visual fidelity and rendering efficiency.
- Established automated, reproducible cross-platform builds and CI pipelines via Nix flakes and GitHub Actions.

Sedmos | *TypeScript, Svelte*

- Engineered a high-performance graphing engine on an interactive canvas, implementing custom world-to-screen coordinate transformations and pan/zoom event handlers for real-time visualization.
- Collaborated with a teammate during a 24-hour hackathon to design responsive UI components, rapidly prototyping and deploying the application under strict time constraints.

ExpenseShare | *ExpressJS, Handlebars, PostgreSQL, NodeJS, Docker*

- Architected a RESTful API backend using Node.js and ExpressJS to manage user authentication, split financial transactions, and process concurrent client requests.
- Designed a normalized PostgreSQL relational database schema to enforce data integrity and secure authorization, enabling efficient CRUD operations for user expenses.
- Collaborated within an Agile team to deliver the application lifecycle under a strict deadline, utilizing Git for version control and Docker for containerized deployment.

Technical Skills

Languages: C++, Rust, Lua, Python, JavaScript, TypeScript, SQL, x86-64 Assembly, Nix

Frameworks & Libraries: React, Svelte/SvelteKit, ExpressJS, NodeJS, Pandas, NumPy, LangGraph

Developer Tools: Git, Docker, AWS, PostgreSQL, Linux, Direnv, OpenAI API

Concepts: Data Science, Agile Methodologies, Frontend, Backend, Systems, Software Engineering